CG5691C Assignment # 1

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1.1 $A = \{x \mid x \in \mathbb{R} \text{ and } -1 \le x \le 2\}$

$B = \{x \mid x \in \mathbb{R} \text{ and } 1 \le x \le 4\}$.

(a) $A \cup B = \{x \mid x \in \mathbb{R} \text{ and } -1 \le x \le 4\}$

(b) $A \cap B = \{x \mid x \in \mathbb{R} \text{ and } 1 \le x \le 2\}$

(c) $A \setminus B = \{x \mid x \in \mathbb{R} \text{ and } -1 \le x \le 1\}$

(d) $\bar{A} \cap B$

$\bar{A} = \{x \mid x \in \mathbb{R} \text{ and } -\infty < x < -1 \text{ \& } 2 < x < \infty)$

$\bar{A} \cap B = \{x \mid x \in \mathbb{R} \text{ and } 2 < x \le 4\}$

Part 2 - Probablity

2.1 Let probablity of getting Heads on coin a

$P(Ha) = 0.6$

similarly, $P(Hb) = 0.4$

(a) Possible Outcomes $= \{HaHb, HaTb, TaHb, TaTb\}$

(b) (i) $E_1 = \{HaHb, HaTb, TaHb\}$

$P(HaHb) = P(Ha) \cdot P(Hb) = 0.24$

$P(HaTb) = P(Ha) \cdot (1 - P(Hb)) = 0.36$

$P(TaHb) = (1 - P(Ha)) \cdot (P(Hb)) = 0.16$

$\therefore P(E_1) = 0.24 + 0.36 + 0.16$

$= 0.76$

(ii) $E_2 = \{ H_a H_b, H_a T_b \}$

$P(H_a H_b) = 0.24$

$P(H_a T_b) = 0.36$

$P(E_2) = 0.24 + 0.36 = 0.6$

(c) $P(H_a H_b) = 0.24$

$P(H_a T_b) = 0.36$

$P(T_a H_b) = 0.16$

$P(T_a T_b) = (1 - P(H_a))(1 - P(H_b)) = 0.24$

2.2 $\Omega = \{ 2.33, 4.56, 5.4, 5.76, 6.32 \}$

$X : \Omega \longrightarrow \Omega_x$

~~$\Omega_x = \{$success, failure$\}$~~

(a) $X(w) : \Omega \longrightarrow \Omega_x$

~~$X(w) = \{$ failure if $w \geq 5$, success else $\}$~~

Observed Probablity of Success $= 2/5 = 0.4$

(as there were only two margins ≤ 5)

X is the number of successful outcomes (trials)

We define $\phi(w_i) = \begin{cases} 0 & \text{if } w_i > 5 \\ 1 & \text{else} \end{cases}$

$X = \sum_{i=1}^{5} \phi(w_i)$ where $w = \{2.33, 4.56, 5.4, 5.76, 6.32\}$

For this case,

$X = 1 + 1 + 0 + 0 + 0$

$= 2$

$\Omega_x = \{0, 1, 2, 3, 4, 5\}$.

(b)

x	$P(X=x)$
0	${}^5C_0(0.4)^0(0.6)^5 = 0.07776$
1	${}^5C_1(0.4)(0.6)^4 = 0.25920$
2	${}^5C_2(0.4)^2(0.6)^3 = 0.34560$
3	${}^5C_3(0.4)^3(0.6)^2 = 0.23040$
4	${}^5C_4(0.4)^4(0.6) = 0.07680$
5	${}^5C_5(0.4)^5(0.6)^0 = 0.01024$

$p = 0.4$ (Probablity for successful trial)

Probablity Mass function $= P(X=K) = \binom{n}{K} p^K (1-p)^{n-K}$

$n=5$ (No of trials) $= \binom{5}{K}(0.4)^K(0.6)^{5-K}$

Part 3 Discrete Random Variable

3.1 $f(K,n,p) = \dfrac{n!}{K!(n-K)!} p^K (1-p)^{n-K}$

$K=45, \quad n=50, \quad p=0.9$

$= \dfrac{50!}{5!\,45!}(0.9)^{45}(0.1)^5 = 0.185$

3.2 $f(K,\lambda) = \dfrac{\lambda^K e^{-\lambda}}{K!} \quad \lambda = 10$

$f(0,\lambda 10) = \dfrac{10^0 e^{-10}}{0!} = e^{-10} \approx 0.0000454$

Part 4

4.1 $P(X=100) = f(100) = \dfrac{10^{100} e^{-10}}{100!}$

$f(k) = \dfrac{10^K e^{-10}}{K!}$

$= 4.864 \times 10^{-63}$

4.2) $f(k) = \frac{10^k e^{-10}}{k!}$

$$P(5 \le x \le 15) = \sum_{k=5}^{15} \frac{10^k e^{-10}}{k!}$$

$$\simeq 0.922$$

4.3) ~~set~~ sample space is set of all non negative integers $\{0, 1, 2, 3 \ldots\}$ with probablity being negligible for very large numbers.

~~4.4~~ 4.5) mean of the observed data =

$$\frac{18 + 20 + 23 + 25 + 22 + 30 + 11 + 16 + 23 + 12}{10}$$

$$= 19.4$$

The expected average number of infections per day according to the researcher's claim = 10

The observed average is nearly double the expected value.

The data suggests that the true generative process likely has a parameter λ close to 19.4, rather than 10. Also 7 out of the observed 10 values have probablity $\approx 2 \times 10^{-7}$

The researcher's model significantly underestimates the number of infections in a day.